

Problem 281: Doomsday

Difficulty: Easy

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Problem Background

Doomsday devices take a LOT of power. Commander Bibbidi Bop wants to supplement LAMBCHOP's quantum antimatter reactor core with solar arrays, and she's tasked you with setting up the solar panels. Due to the nature of the space station's outer paneling, all of its solar panels must be squares. Fortunately, you have one very large and flat area of solar material, a pair of industrial-strength scissors, and enough MegaCorp Solar Tape™ to piece together any excess panel material into more squares. For example, if you had a total area of 12 square yards of solar material, you would be able to make one 3x3 square panel (with a total area of 9). That would leave 3 square yards, so you can turn those into three 1x1 square solar panels.

Problem Description

Write a program that takes as its input a single unit of measure representing the total area of solar panels you have and returns a list of the areas of the largest squares you could make out of those panels, starting with the largest squares first. So, following the example above, 12 would return [9, 1, 1, 1]. Each square must have integer-length sides.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include an integer between 1 and 1,000,000 inclusive that represents the total area of solar panels that you have.

```
2
12
15324
```

Sample Output

For each test case, your program must print a list of areas of the largest squares that can be made from the provided area, in order from largest to smallest. The areas must be in a list with an opening and closing square brackets surrounding the areas, which are listed with a comma and space between each of them. The last area listed should not have a comma or space after it.

```
[9, 1, 1, 1]
[15129, 169, 25, 1]
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